

caRNA-seq 10X Genomics Sample Prep Protocol

Objective

This protocol allows for the detection of single-cell gene expression and CAR-specific barcodes from a single 10X sample (5' chemistry) via target-specific amplification from the full-length cDNA library following cDNA amplification and clean-up. Target-specific enrichment of the CAR barcodes is achieved with a nested PCR strategy, largely similar to the 10X V(D)J enrichment protocol.

Materials

- All Chromium Next GEM Single Cell 5' v2 reagents, with the exception of V(D)J amplification oligos, which are not required here
- 2X KAPA HiFi HotStart ReadyMix
- SPRISelect Beads
- CAR barcode amplification oligos, 100 uM stocks

Primer Name	Full Sequence (5' to 3')	Description
10X_Univ_F	GATCTACACTCTTTCCCTACACGACGC	F primer for PCR1 and 2; anneals to 10X Read1 handle; includes handle for 10X SI-PCR
10X_CARBC_PCR1_R	GCGCTAAGACTGGAAGTTGTTTGC	R primer for PCR1; anneals to CD19 scFv
10X_CARBC_PCR2_R	GTGACTGGAGTTCAGACGTGTGCTCTTC CGATCTGCAAAGCCATGGTGGCTCTAGA	R primer for PCR2; anneals just 3' of CAR barcode; includes handle for 10X SI-PCR

Protocol

GEM Generation and cDNA Amplification:

1. Follow the [10X Chromium Next GEM Single Cell 5' v2 User Guide](#) as written through Step 2 (cDNA amplification), loading each sample into a separate lane of a 10X chip. **Each sample is to be handled separately throughout this protocol, only to be pooled for NGS following separate sample indexing.**
2. Following SPRI clean-up of amplified cDNA for both samples as described in the 10X User Guide, retain all 45 uL of the large fraction, which will be used for both CAR barcode amplification and gene expression library construction.

CAR Barcode Nested PCR Amplification:

3. Perform CAR barcode PCR1 as follows:
 - a. Prepare PCR1 reaction mix for each sample. **Be sure to use PCR1 R primer.**

Reagent	Volume for 1X Reaction	Final Concentration
2X KAPA HiFi HotStart ReadyMix	25 uL	1X
10 uM 10X_Univ_F Primer	1 uL	0.2 uM
10 uM CAR_BC_PCR1_R	1 uL	0.2 uM
Amplified cDNA	1 uL	---
Nuclease-free water	22 uL	---
Total	50 uL	

- b. Load thermocycler with the following cycling conditions (**10 total cycles**).

Step	Temperature	Time
1	98°C	45 sec
2	98°C	20 sec
3	65°C	30 sec
4	72°C	30 sec
5	Go to step 2, repeat 9x	
6	72°C	1 min
7	4°C	Hold

4. Perform two-sided 0.5/1X SPRI selection to enrich for desired product of PCR1 (**expected amplicon size of 285 bp**), as follows:
- Add 25 uL of SPRISelect beads to each 50 uL PCR product (**0.5X SPRI**), mix, and incubate as indicated in SPRISelect protocol.
 - Following incubation of magnet, transfer 70 uL of supernatant from each sample to a new PCR tube. **DO NOT DISCARD SUPERNATANT.**
 - Add an additional 23.3 uL of SPRISelect beads to the 70 uL of supernatant (23.3 uL SPRISelect + 46.6 uL sample) to achieve a **1X SPRI** ratio. Mix, incubate, and wash as indicated in the SPRISelect protocol.
 - Elute enriched amplicon off of the beads in 25 uL of water.
5. Perform CAR barcode PCR2 as follows:
- Prepare PCR2 reaction mix for each sample. **Be sure to use PCR2 R primer.**

Reagent	Volume for 1X Reaction	Final Concentration
2X KAPA HiFi HotStart ReadyMix	25 uL	1X
10 uM 10X Univ F Primer	1 uL	0.2 uM
10 uM CAR BC PCR2_R	1 uL	0.2 uM
Eluted PCR1 product	23 uL	---
Total	50 uL	

- b. Load thermocycler with the following cycling conditions (**15 total cycles**).

Step	Temperature	Time
1	98°C	45 sec
2	98°C	20 sec
3	65°C	30 sec
4	72°C	30 sec
5	Go to step 2, repeat 14x	
6	72°C	1 min
7	4°C	Hold

6. To clean PCR2 product (**expected amplicon size of 204 bp**) and remove primer dimers, perform left-sided **1.2X SPRI selection** by adding 60 uL of SPRISelect beads to 50 uL of PCR2 product. Mix, incubate, and wash as described in SPRISelect protocol. Elute cleaned amplicon off of beads in 35 uL of water.
7. Confirm size and purity of PCR2 product by fragment analyzer before proceeding to sample indexing.

Endogenous Gene Expression Library Construction:

- Construct gene expression library as normal, as described in the Step 5 of the [10X Chromium Next GEM Single Cell 5' v2 User Guide](#).
- Note that sample indexing can be done in parallel with the CAR barcode amplicon library, as described below.

CAR Barcode and Gene Expression Library Sample Indexing:

- The CAR barcode amplicons generated in PCR2 contain the same primer handles as 10X gene expression or V(D)J libraries, allowing the use of 10X-compatible sample indexing primers for indexing both CAR barcode amplicons and gene expression libraries.
- Prior to prepping reactions, ensure distinct sample indexes are selected for CAR barcodes and gene expression libraries, as well as different samples (i.e. for 2 samples, we need 4 distinct indexes—2 for each gene expression library, and 2 for each CAR barcode library).
- Prepare gene expression sample indexing PCR reactions as normal, as described in the 10X User Guide.
- Prepare CAR barcode sample indexing PCR reactions exactly as described in the User Guide, **replacing the 30 uL of gene expression library with 10 ng of cleaned CAR barcode PCR2 product (diluted into 30 uL with water).**
- Load the all reactions into a thermocycler, following the exact cycle conditions described in the 10X User Guide; **however, perform only 7 total cycles for the amplicon library.**
- Clean the gene expression library reactions as described in the 10X User Guide.
- Clean the CAR barcode library reactions with a left-sided **1.2X SPRI. This is similar to the clean-up used for 10X Feature Barcoding libraries (Step 6.2 in the User Guide, page 68), and this exact protocol can be followed.**
- Perform QC of final gene expression and CAR barcode libraries. The expected final amplicon size of the CAR barcode libraries following sample indexing is **272 bp**.

Sequencing:

- Pool indexed samples such that **CAR barcode libraries make up 10% of the total reads.** See example pooling ratios below:

Sample	Index Name	Pooled Fraction
Sample1 GEX	SI-1	22.5%
Sample1 CAR BC	SI-2	2.5%
Sample2 GEX	SI-3	22.5%
Sample2 CAR BC	SI-4	2.5%
Sample3 GEX	SI-5	22.5%
Sample3 CAR BC	SI-6	2.5%
Sample4 GEX	SI-7	22.5%
Sample4 CAR BC	SI-8	2.5%

- Both gene expression libraries and CAR barcode libraries should include standard Illumina TruSeq Read1 and Read2 priming sites. Therefore, the resulting pool can be sequenced in a single NextSeq or NovaSeq run using normal 10X gene expression sequencing primers and parameters.
- Select a sequencer based on the total number of input cells (see below table), aiming for a read depth of at least 22.2K reads/cell (20K gene expression reads/cell, assuming gene expression library is pooled at 90%).

Sequencer	Default Read Length	Avg. Number of Reads	Number of Cells	Cost
NextSeq / NovaSeq 400M	28.8.55 / 28.8.288	400M	20K	\$2270 / \$1600
NovaSeq SP	28.8.88	800M	40K	\$3365
NovaSeq S1	28.8.88	1.6B	80K	\$5300
NovaSeq S4 (single lane)	28.8.288	2B	100K	\$5000

- a. Note that for runs pooled across multiple projects (NovaSeq 400M or NovaSeq S4 single lane), you get greater value per read, but may have to wait for other projects to get submitted before running. These cost values reflect the MIT BioMicroCenter Core pricing, as of mid-2022; although costs have likely risen since, the relative value across them are likely similar.